

Esercizi termodinamico

1) $L = 301 \text{ m}$ $\alpha_{Fe} = 12 \cdot 10^{-6} \text{ } ^\circ\text{C}^{-1}$

$$T_i = 220^\circ\text{C}$$

$$T_f = 0^\circ\text{C}$$

$$\Delta L = L \alpha \Delta T = L \alpha (-220) = -0.8 \text{ m}$$

$$\Delta L = L_f - L_i$$

$$L_f = \Delta L + L_i = 300.2 \text{ m}$$

2) $m_g = 0.7 \text{ kg}$
 $T_g = -10^\circ\text{C}$
 $m_t = 3 \text{ kg}$
 $T_t = 15^\circ\text{C}$

$T_f?$

$$L_f = 333 \text{ kJ/kg}$$

$$C_e = 4186 \text{ J/kg } ^\circ\text{C}$$

$$C_g = 2100 \text{ J/kg } ^\circ\text{C}$$

Per stabilire a che temperatura e quale sarà la miscela finale verifichiamo se il ghiaccio si è sciolto tutto.

$$Q_g = m C_g \Delta T = m C_g (0 - (-10)) = 14.7 \text{ kJ}$$

$$Q_{gf} = m L_f = 233.1 \text{ kJ}$$

$$Q_{\text{tot}} = Q_g + Q_{gf} = 247.8 \text{ kJ}$$

$$Q_t = m C_e \Delta T = m C_e (0 - 15) = -188.4 \text{ kJ}$$

Il calore fornito dal tè non è sufficiente a fondere tutto il ghiaccio:

$$Q_g + Q_{gh} > |Q_t| \Rightarrow T_f = 0^\circ \text{C}$$

Calcoliamo quanto ghiaccio si è sciolto.

$$Q_g + Q_{gf} + Q_t = 0$$

$$mL_f = -Q_g - Q_t \Rightarrow m = \frac{(-14.7 + 138.4) \text{ kJ}}{333 \text{ kJ/kg}} = 0.52 \text{ kg}$$

3) $n = 3 \text{ mol}$
 $p_i = 3 \cdot 10^5 \text{ Pa}$
 $T = 300^\circ \text{K}$ (costante)
 $p_f = 10^5 \text{ Pa}$

$\Delta T?$
 $\Delta E_{\text{int}}?$
 $L?$
 $Q?$

$$pV = nRT$$

$$V_i = \frac{nRT}{p_i} = 0.025 \text{ m}^3$$

$$V_f = \frac{nRT}{p_f} = 0.075 \text{ m}^3$$

$$\Delta V = V_f - V_i = 0.05 \text{ m}^3$$

$$\Delta E_{\text{int}} = nC_v \Delta T = nC_v (300 - 300) = 0$$

$$L = nRT \ln\left(\frac{V_f}{V_i}\right) = 3216.5 \text{ J}$$

$$Q = L = 3216.5 \text{ J}$$

4) $n = 1 \text{ mol}$
 $T_A = 300^\circ \text{K}$
 $T_B = 600^\circ \text{K}$
 $p_A = 10^5 \text{ Pa}$
 $T_C = 455^\circ \text{K}$
 $V_C = 0.038 \text{ m}^3$

$A(p_A, T_A, V_A)?$
 $B(p_B, T_B, V_B)?$
 $C(p_C, T_C, V_C)?$
 $L_{AB}, L_{BC}, L_{CA}, L_{\text{TOT}}?$
 $Q_{AB}, Q_{BC}, Q_{CA}, Q_{\text{TOT}}?$

$$A(10^5 \text{ Pa}, 300^\circ \text{K}, V_A)$$

$$V_A = \frac{nR\bar{T}_A}{P_A} = 0.025 \text{ m}^3 \Rightarrow A(10^5 \text{ Pa}, 300^\circ \text{K}, 0.025 \text{ m}^3)$$

$$B(P_B, 600^\circ \text{K}, V_B)$$

$$C(P_C, 455^\circ \text{K}, 0.038 \text{ m}^3)$$

La trasformazione CA è isobara, e pressione costante, quindi

$$P_A = P_C \Rightarrow C(10^5 \text{ Pa}, 455^\circ \text{K}, 0.038 \text{ m}^3)$$

La trasformazione AB è isocora, e volume costante, quindi

$$V_B = V_A \Rightarrow B(P_B, 600^\circ \text{K}, 0.025 \text{ m}^3)$$

$$P_B = \frac{nR\bar{T}_B}{V_B} \approx 2 \cdot 10^5 \text{ Pa}$$

AB è isocora, V è costante $\Rightarrow L_{AB} = 0$

$$Q_{AB} = n C_V \Delta T = n \frac{3}{2} R (\bar{T}_B - \bar{T}_A) = 3739.5 \text{ J}$$

BC è adiabatica $\Rightarrow Q_{BC} = 0$

$$L_{BC} = -n C_V \Delta T = -n C_V (\bar{T}_C - \bar{T}_B) = 1807.4 \text{ J}$$

CA è isobara, e p costante: $L_{CA} = p \Delta V = p \frac{nR \Delta \bar{T}}{p} = nR \Delta \bar{T} = nR(\bar{T}_A - \bar{T}_C) = -1288 \text{ J}$

$$Q_{CA} = n C_P \Delta T = n \frac{5}{2} R (\bar{T}_A - \bar{T}_C) = -3220.1 \text{ J}$$

In un ciclo $\Delta E_{\text{int, tot}} = 0 \Rightarrow Q_{\text{tot}} = L_{\text{tot}}$

$$Q_{\text{tot}} = Q_{AB} + Q_{BC} + Q_{CA} = (3739.5 + 0 - 3220.1) \text{ J} = 519.4 \text{ J}$$

$$\Sigma) m_{\text{cu}} = 300 \text{ g} = 0.3 \text{ kg}$$

$$T_{\text{in, cu}} = 90^\circ \text{C}$$

$$m_{\text{acqua}} = 0.7 \text{ kg}$$

$T_{\text{eq}}?$

$$m_{Al} = 700 \text{ g} = 0.7 \text{ kg}$$

$$T_{i, Al} = 43^\circ \text{C}$$

$$C_{Cu} = 389 \text{ J/kg}^\circ \text{C} \quad C_{Al} = 900 \text{ J/kg}^\circ \text{C}$$

$$Q_{Cu} = m C_{Cu} \Delta T = m C_{Cu} (T_{eq} - T_i) = m C_{Cu} T_{eq} - m C_{Cu} T_i = m C_{Cu} T_{eq} - 10503 \text{ J}$$

$$Q_{Al} = m C_{Al} \Delta T = m C_{Al} (T_{eq} - T_i) = m C_{Al} T_{eq} - m C_{Al} T_i = m C_{Al} T_{eq} - 27090 \text{ J}$$

Se il sistema è in equilibrio $\Rightarrow Q_{tot} = 0$

$$Q_{Cu} + Q_{Al} = 0$$

$$m C_{Cu} T_{eq} - 10503 \text{ J} + m C_{Al} T_{eq} - 27090 \text{ J} = 0$$

$$T_{eq} (m_{Cu} C_{Cu} + m_{Al} C_{Al}) = 37593 \text{ J}$$

$$T_{eq} (116.7 \text{ J} + 630 \text{ J}) = 37593 \text{ J}$$

$$T_{eq} = 50.3^\circ \text{C}$$

6) $m_g = 1.5 \text{ kg}$
 $T_{ig} = -40^\circ \text{C}$
 $m_e = 2 \text{ kg}$
 $T_{ie} = 20^\circ \text{C}$

$m_g \text{ fuse?}$

$$L_f = 333 \text{ kJ/kg}$$

$$c_e = 4186 \text{ J/kg}^\circ \text{C}$$

$$c_g = 2093 \text{ J/kg}^\circ \text{C}$$

Il calore necessario al ghiaccio per fondersi è:

$$Q_g = m_g c_g \Delta T = m_g c_g (0 - (-40)) = 125.6 \text{ kJ}$$

$$Q_{gf} = m_g L_f = 499.5 \text{ kJ}$$

$$Q_{g \text{ tot}} = 625.1 \text{ kJ}$$

Il calore fornito dall'acqua è:

$$Q_e = m_e c_e \Delta T = m_e c_e (0 - 20) = -167.4 \text{ kJ}$$

$Q_{\text{gl}} > |Q_e| \Rightarrow$ Il ghiaccio non si fonde tutto quindi $T_f = 0$

Calcoliamo quante masse di ghiaccio si è fuse:

$$Q_g + Q_{gf} + Q_t = 0$$

$$Q_{gf} = -Q_g - Q_t \Rightarrow m_g L_f = -Q_g - Q_t$$

$$m_g = \frac{-Q_g - Q_t}{L_f} = 0.13 \text{ kg}$$

1) $q_1 = 8q$ in $x=0$

$$q_2 = -2q \text{ in } x=L$$

$$p \text{ in } x=?$$

Affinché p sia in equilibrio $\sum_{i=1}^n \vec{F}_i = 0$

$$F_1 + F_2 = 0$$

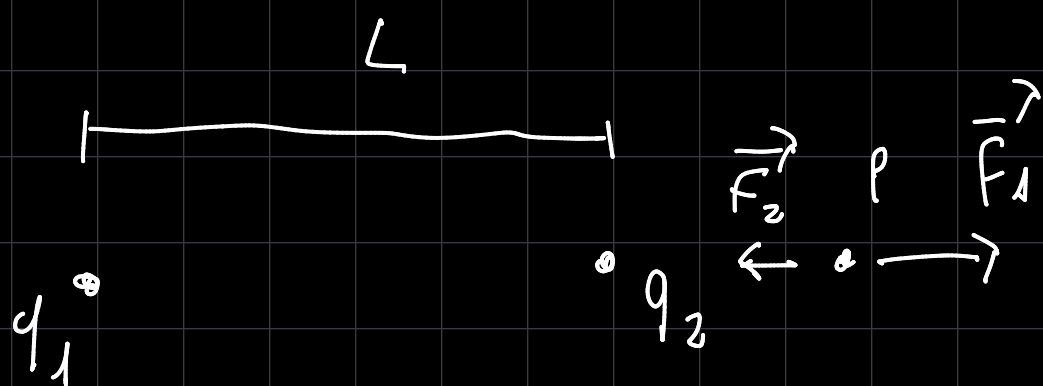
$$k \frac{q_1 q}{x^2} + k \frac{q_2 q}{(x-L)^2} = 0$$

$$\frac{8q}{x^2} = \frac{2q}{(x-L)^2}$$

$$4(x-L)^2 = x^2$$

$$2(x-L) = x$$

$$2x - 2L = x \Rightarrow x = 2L \Rightarrow x = 22$$



$$\vec{F}_1 \nearrow$$

7 nC

3) $d = 0.5 \text{ m}$
 $\theta = 60^\circ$

$$F_1 = \frac{k (7 \cdot 10^{-6} \text{ C})(2 \cdot 10^{-6} \text{ C})}{(0.5)^2} = 0.503 \text{ N}$$

$$F_2 = k \frac{(7 \cdot 10^{-6} \text{ C})(4 \cdot 10^{-6} \text{ C})}{(0.5)^2} = 1.01 \text{ N}$$

$$F_x = 0.503 \cdot \cos 60 + 1.01 \cdot \cos 60 = 0.755 \text{ N}$$

$$F_y = 0.503 \cdot \sin 60 + 1.01 \cdot \sin 60 = -0.436 \text{ N}$$

$$F = \sqrt{F_x^2 + F_y^2} = 0.872$$

$$\theta_F = \tan^{-1} \left(\frac{F_y}{F_x} \right) = 300^\circ$$

